

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 14

(1) Research Program/Project/Study Title: Performance Of Indigenous Rice Applied with Bokashi with Different Manures

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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check)

- ☐ Social Science/Education /Humanities & other related fields
☐ Natural/Biological Sciences
☐ Food Research/
☐ Developmental/ Computer/Engineering/ Architecture
☐ Extension
☐ Poster

(3) Research Center

[Check the concerned center]

- ☐ SSITRDC
 Social Science and Industrial
 Technology Research
 Development Center

☐ LRDC
 Livestock Research
 Development Center

☐ FFRDC
 Fisheries and Food Research
 Development Center

☒ CSRDC
 Crop Science Research
 Development Center

(4) Site of implementation

Municipality/City:

Province:

Region:

(5) Classification

Research:

Development:

☐ Basic

☐ Pilot Testing

☐ Applied

☐ Tech. Promotion/Commercialization

(6) Mode of Implementation

- ☐ Single Authorship/Proponent
☐ Collaborative Research (within campus)
☐ Collaborative Research (intercampus)
☐ Collaborative Research

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 14

		(external agency)
(7) Sector/Commodity		(8) Discipline/Field of Specialization
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) <i>This study was conducted to evaluate the performance of indigenous rice applied with bokashi with different farm manure. It was laid out using the Randomized Complete Block Design (RCBD) with two factors. Factor A consisted bokashi with different farm manures such as: A1- bokashi with chicken dung, A2 – bokashi with sheep manure, A3 – hog manure, and A4 - Nutriboost (Control) . On the other hand, Factor B comprised the different varieties of indigenous rice, namely: B1 – Red rice, B2 – Black rice, B3 – Kaponpon rice, and B4 – Japanese rice. Each treatment was replicated three times. All data gathered were analyzed through the IRRISTAR Program and results were interpreted at 5% level of significance. Differences among treatments was determined using the Duncan Multiple Range Test (DMRT) and were interpreted at 5% level of significance.</i> <i>All data parameters was taken from ten (10) sample plants.</i> <i>The results revealed that growth and yield of the different indigenous rice, when applied with bokashi with different farm manures, varies significantly.</i> <i>The application bokashi with different farm manures gave similar results in all the growth and yield parameters gathered.</i> <i>Interaction effect was noted on plant height, biomass, number of productive tillers, number of filled grains, number of unfilled grains, number of spikelets per panicle.</i> (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) <i>Keywords: bokashi, manure, indigenous rice, nutriboost</i>		
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) . Rice is labeled by the National Geographic Encyclopedia (2022) as a staple food consumed by more than 3.5 billion people worldwide, particularly in Asia, Latin America, and parts of Africa. Like almost all Asian countries, the Philippines has cultivated rice for thousands of years. Rice, commonly known as "Palay" (unmilled rice), "Bigas" (milled rice), and "Kanin" (cooked rice), is much more than food in the Philippines (National Nutrition Council, 2020). It is a grain crop that is significant in terms of history, culture, agriculture, and the economy. It is one of the major contributing crops to the Philippines' economy. It contributed 14% of the country's gross domestic product and 13 million workers (32 percent) comprise the labor force as of 2012. The Philippine government strongly supports increasing rice production because it is one of the top rice-producing nations. Organic farming is not new to Philippine agriculture, but it is still on its infancy stage. Organic rice growing is one of the improvements that the government is promoting as it is deemed as safer and low cost. Organic farming is both beneficial to the farmers and the environment. In the research paper conducted by Ara (2002) it was mentioned that farmers can		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 14

benefit in two ways from organic agriculture; first, is the low production cost and second, farmers can improve the fertility of their land, environment, biodiversity, soil and water quality and their own health condition.

Ancient farmers have utilized organic fertilizers—fertilizers that come from byproducts or results of naturally occurring processes—for thousands of years. Today, organic farming is becoming more and more popular as a sustainable agricultural method (Singh, 2012). One kind of organic fertilizer used by farmers today is the bokashi. Bokashi is similar to compost, which is the end result of the microbial breakdown of organic waste like manure, soil, and agricultural leftovers high in carbon like rice hulls and bran. The distinction between bokashi and compost is that bokashi processing uses both aerobic and anaerobic microbial processes. Organic matter in bokashi breaks down more quickly due to the partially anaerobic conditions and energy-dense organic stuff present. Bokashi matures in about two weeks and is made in smaller piles for easier maintenance, whereas composting frequently necessitates lengthy maturation durations, constant aeration and hydration, and enormous expanses (Kittredge, 2022).

Despite the efforts of the Philippine government to promote organic rice farming, the adoption rate is still low due to some circumstances. Pantoja et al. (2016) identified why farmers refuse to adopt organic farming. It was stated that farmers' see organic rice production as more laborious and time-consuming compared to conventional practices and that this type of farming declines their yield. These were aggravated by production problems (e.g., pests and diseases, natural calamities, and lack of irrigation water), lack of workers competent in organic farming, limitations on access to organic inputs such as fertilizer and seeds, and delay in the delivery of inputs.

ECO gardener (2018) also discussed the advantages and disadvantages of organic and chemical fertilizers which is also one of the things being considered by farmers. According to them, using organic fertilizers improves the overall nutrient profile of the soil while also improving its structure, slow release of nutrients that allows plants to absorb the nutrients they need to grow without stressing out the crop, no toxin buildup, 100% eco-friendly, and you can create your organic fertilizer. Organic fertilizer, like everything else, has drawbacks; the effect is limited due to the slow breakdown of organic fertilizers, and farmers will have a slow start. Chemical fertilizers, on the other hand, are less expensive, produce faster results, and reduce the risk of over-fertilization; however, these fertilizers are not eco-friendly, do not improve soil structure, require frequent application, and cause toxic buildup.

With this, the researcher would like to see if using bokashi with chicken, hog and sheep manure improves the performance or growth of the indigenous rice compared to the nutriboost fertilizer.

Objective of the Study

The study was conducted to determine the performance of indigenous rice varieties applied with bokashi with different farm manures. Specifically, the study aims to;

1. determine the growth and yield performance of indigenous rice varieties of indigenous rice applied with bokashi with different farm manures;
2. best substrates in growing different indigenous rice varieties;
3. interaction effect between the different rice varieties and bokashi with different farm manures.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 14

This study was conducted to evaluate the performance of indigenous rice applied with bokashi with different farm manure. It was laid out using the Randomized Complete Block Design (RCBD) with two factors. Factor A consisted bokashi with different farm manures such as: A1- bokashi with chicken dung, A2 – bokashi with sheep manure, A3 – hog manure, and A4 - Nutriboost (Control) . On the other hand, Factor B comprised the different varieties of indigenous rice, namely: B1 – Red rice, B2 – Black rice, B3 – Kaponpon rice, and B4 – Japanese rice. Each treatment was replicated three times. All data gathered were analyzed through the IRRISTAR Program and results were interpreted at 5% level of significance. Differences among treatments was determined using the Duncan Multiple Range Test (DMRT) and were interpreted at 5% level of significance.

All data gathered was taken from ten (10) sample plants. Following

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

RESULTS AND DISCUSSION

Plant Height (cm)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	137.62 ^{ab}	144.95 ^a	135.94 ^b	144.99 ^a	563.5	140.88
B- Sheep Manure	132.75 ^b	143.91 ^a	150.27 ^a	149.66 ^a	567.59	141.9
C- Hog Manure	141.05 ^a	145.73 ^a	141.12 ^a	144.10 ^a	572	143
D- Nutriboost (control)	136.13 ^{ab}	143.40 ^a	131.29 ^b	142.31 ^a	553.13	138.28
TOTAL	547.55	577.99	558.62	581.06		
MEAN	136.89^c	144.50^{ab}	139.66^{bc}	145.27^a		

The ANOVA for the plant height revealed a highly significant result for indigenous rice varieties and a significant effect of bokashi with different farm manures.

There was also an interaction effect between the varieties of the different indigenous rice and bokashi with different farm manures.

The Kaponpon variety, fertilized with bokashi with sheep manure, has the tallest plant height with a mean of 150.27 cm, which is found to be comparable with the plant height of Japanese rice (149.66 cm) and black rice(143.91 cm) fertilized with bokashi with sheep manure, Japanese rice(144.99 cm), and black rice (144.95) fertilized with bokashi with chicken dung, black rice(145.73 cm), Japanese rice(144.10 cm), kaponpon (141.12 cm), and red rice (141.05) when fertilized with hog manure and black rice (143.40 cm). Red rice (136.13 cm) was applied with nutriboost. Kaponpon obtained the shortest plant height applied nutriboost with a mean of 131.29 cm, followed by kaponpon (135.94 cm) applied with bokashi with chicken dung, and red rice (132.75 cm) applied with bokashi with sheep manure.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
				Revision No.	00
	Document Title:	COMPLETED RESEARCH		Effective Date	
				Page	Page 5 of 14

Biomass (grams)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	158.93a	127.80a	131.90a	137.56a	556.19	139.05
B- Sheep Manure	164b	140.43b	215.83a	148.26b	668.52	167.13
C- Hog Manure	172.83a	164.13a	181.30a	119.50b	637.76	159.44
D- Nutriboost (control)	148.33a	143.23ab	154.66a	111.10b	557.32	139.33
TOTAL	644.09	575.59	683.69	516.42		
MEAN	161.02	143.90	170.92	129.11		

The biomass of different varieties of indigenous rice was highly significantly affected both by the application of bokashi with different farm manures and the varieties. A positive interaction effect between the two factors was also noted.

Among the different indigenous rice varieties, the analysis resulted to the Red Rice having the highest biomass with a mean of 170.92g where the highest biomass amongst the treatments can also be found; kaponpon fertilized with sheep manure with a mean of 215.83g. On the other hand, the Japanese Rice resulted to the lowest biomass with a mean of 129.11g.

In terms of the bokashi substrates, table showed that Sheep manure produces the highest biomass with a mean of 167.13g followed by hog manure (159.44g), nutriboost(139.33g) and chicken dung (139.05g).

No. of Leaves

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	24.86	21.63	16.83	13.46	76.78	19.2
B- Sheep Manure	27.4	22.76	22.8	14	86.96	21.74
C- Hog Manure	26.9	27.03	21.63	13.53	88.19	22.05
D- Nutriboost (control)	26.63	23.4	18.23	11.56	79.82	20
TOTAL	105.79	94.82	79.49	52.55		
MEAN	26.45a	23.71a	19.87b	13.14c		

The analysis revealed a highly significant effect on different indigenous rice varieties but not on Bokashi with different farm manures in terms of the number of leaves. On different indigenous rice varieties, Red rice variety has the most number of leaves with the mean of 26.45 leaves but was comparable to black rice variety, while using Japanese variety was recorded with the lowest number of leaves with the mean of 13.14 leaves. On the other hand, bokashi with different farm manures ranged from 22.05 to 19.20 number of leaves.

Moreover, there was not a significant difference between the different indigenous rice varieties and bokashi with different farm manures.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 6 of 14

Stem Girth (mm)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	24.5	21.23	27.2	26.56	99.49	24.87
B- Sheep Manure	25.06	21.43	28.36	26.53	101.38	25.35
C- Hog Manure	25.46	20.96	26	26.76	99.18	24.8
D- Nutriboost (control)	25.73	21.7	27.36	26.36	101.15	25.29
TOTAL	100.75	85.32	108.92	106.21		
MEAN	25.19b	21.33c	27.23a	26.55ab		

Analysis revealed that, there was a significant effect on the different indigenous rice varieties but not in the bokashi with different farm manures. In terms of different indigenous rice varieties, Kaponpon variety was observed to have the thickest stem girth with the mean of 27.23 mm, while Black rice variety, with the mean of 21.33 mm, has the thinnest width. In terms of bokashi with different farm manures, mean ranges from 25.35 mm to 24.80 mm.

However, there was no significant difference between the different indigenous rice varieties and bokashi with different farm manures

No. of Tillers

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	11.03	9.1	4.96	5.86	30.95	7.74
B- Sheep Manure	10.7	9.8	7.73	5.9	34.13	8.53
C- Hog Manure	10.46	11.43	7.6	4.93	33.42	8.36
D- Nutriboost (control)	11	10.23	7.03	4.5	32.76	8.19
TOTAL	43.19	40.56	27.32	21.19		
MEAN	10.80a	10.14a	6.83b	5.30c		

The analysis of variance for the number of tillers revealed that there was a significant effect on the different indigenous rice varieties but not on bokashi with different farm manures. In terms of bokashi with different farm manures, mean ranges from 8.53 to 7.74 tillers. In terms of different indigenous rice varieties, red rice has the most number of tillers with the mean of 10.80 but was found to be comparable to black rice, while the least number of tillers was obtained from Japanese variety with the mean of 5.30 tillers.

However, there was no significant difference between different indigenous rice varieties and bokashi with different farm manures.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 14

No. of Productive Tillers

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	9.73a	8.23a	3.60c	5.26b	21.56	5.39
B- Sheep Manure	9.96a	8.80a	6.30b	5.10b	30.16	7.54
C- Hog Manure	9.86a	10.50a	6.96b	4.33c	31.65	7.91
D- Nutriboost (control)	10.56a	9.40a	5.26b	3.90b	29.12	7.28
TOTAL	40.11	36.93	22.12	18.59		
MEAN	10.03	9.23	5.53	4.65		

The analysis of variance for the number of productive tillers revealed a significant result on the effect of bokashi made from different farm manures and a highly significant effect on different indigenous rice varieties. An interaction effect between the two factors was also reflected in the result of ANOVA. This implies that the two factors influence the number of productive tillers.

In terms of different indigenous varieties, the mean ranged from 10.03 to 4.65 number of tillers, while on bokashi with different farm manure, the mean ranged from 7.91 to 5.39 number of tillers.

Moreover, there was a significant interaction between different indigenous varieties and bokashi with different farm manure. It is found out that using nutriboost when planting red rice recorded the most productive tillers.

No. of Non-productive Tillers

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	1.26	0.86	1.36	0.76	4.24	1.06
B- Sheep Manure	0.73	1.23	1.4	0.8	4.16	1.04
C- Hog Manure	0.6	0.96	0.63	0.6	2.79	0.7
D- Nutriboost (control)	0.43	0.5	1.76	0.3	2.99	0.75
TOTAL	3.02	3.55	5.15	2.46		
MEAN	0.76b	0.89b	1.29a	0.62c		

Analysis revealed that the number of non-productive tillers was significantly affected by the indigenous rice varieties but not in bokashi with different farm manure. In terms of bokashi with different farm manure, the mean ranged from 1.06 to 0.70 number of tillers. On the other hand, Kaponpon variety has the most number of non-productive tillers, with a mean of 1.29 of non-productive tillers. In contrast, the lowest number of non-productive tillers was found in the Japanese variety with a mean of 0.62 of non-productive tillers.

However, there was no significant interaction between indigenous rice varieties and bokashi with different farm manures.

Panicle Length (cm)

BOKASHI	RICE VARIETIES	TOTAL	MEAN
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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 8 of 14

SUBSTRATES	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	23.09	21	25.85	26	95.94	23.99
B- Sheep Manure	22.96	21.14	28.07	27.09	99.26	24.82
C- Hog Manure	21.99	21.92	26.93	28.63	99.47	24.87
D- Nutriboost (control)	22.44	22.77	26.61	26.55	98.37	24.59
TOTAL	90.48	86.83	107.46	108.27		
MEAN	22.62	21.71	26.87	27.07		

Analysis revealed that the panicle length was not significantly affected by the different indigenous rice varieties and bokashi with different farm manures. In terms of different indigenous rice varieties, the mean ranged from 21.71 cm to 27.07 cm in length. In terms of bokashi with different farm manures, the mean ranged from 23.99 cm to 24.87 cm in length.

Moreover, there was no significant difference between different indigenous rice varieties and bokashi with different farm manures.

Weight of Panicle (g)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	4.16	3.37	5.27	4.85	17.65	4.41
B- Sheep Manure	3.9	3.24	4.86	4.68	16.68	4.17
C- Hog Manure	3.73	3.25	4.93	5.66	17.57	4.39
D- Nutriboost (control)	3.77	3.79	4.09	4.56	16.21	4.05
TOTAL	23.88	13.65	19.15	19.75		
MEAN	5.97	3.41	4.79	4.94		

The ANOVA for the weight of the panicle revealed a highly significant result on the effect of different rice varieties and a not significant result for bokashi with the different farm manures. Japanese rice (4.94) grams has the heaviest panicle, which is comparable with that of Kaponpon rice (4.79) g, while black rice (3.41) g has the lightest panicle with an equal panicle weight to red rice (3.90) g.

No interaction between the two factors was obtained based on the results of ANOVA.

Number of Filled Grains

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	104.23 b	103.37 b	235.77 a	107.7 b	551.07	137.77
B- Sheep Manure	96.87 c	98.63 c	199.5 a	125.57 b	520.57	130.14
C- Hog Manure	114.33 c	97.13 c	194.77 a	141.7 b	547.93	136.98
D- Nutriboost (control)	87.17 c	104 ac	169.23 a	115.47 b	475.87	118.97
TOTAL	402.6	403.13	799.27	490.44		
MEAN	100.65c	100.78c	199.82a	122.61b		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 14

The analysis for the number of filled grains for the number of filled grain revealed that there was a significant effect on the different indigenous rice varieties but not on bokashi with different farm manures. In terms of bokashi with different frm manures, the mean ranged from 130.14 to 137.77 number of filled grains. In terms of different indigenious rice varieties, the highest mean was obtained from Kaponpon varieties with the mean of 199.82 number of grains, while the least number of grains was obtained from Red rice variety with the mean of 100.65 numberof grains.

Moreover, there was a significant difference between differen indigenious rice varieties and bokashi with different farm manures. Using Bokashi with chicken dung on Kaponpon variety yielded the most number of filled grains with a mean of 235.77 grains, but this was comprabl to the same variety using bokashi with sheep manure, hog manure and nutriboost, with the means of 199.5, 194.77 and 169.23 grains, respectively. The least number of filled grains was obtained from Red rice variety nutriboost.

Number of Unfilled Grains

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanes e		
A- Chicken Dung	21.97b	24.23b	56.5a	46.53b	149.23	37.41
B- Sheep Manure	21.13c	29.07c	111.47a	54.67a	216.34	54.09
C- Hog Manure	17.43b	38.5b	79.7a	36.23b	171.86	42.97
D- Nutriboost (control)	28.03b	28.63b	81.43a	40.73b	178.82	44.71
TOTAL	88.56	120.43	329.1	178.16		
MEAN	22.14c	30.11c	82.28a	44.54b		

The ANOVA for the number of unfilled grains revealed a highly significant effect on the different rice varieties. In contrast, the effect of bokashi with different farm manures is revealed to be not significant. The mean ranged from 37.41 to 54.09. For the different rice varieties, Kaponpon rice (82.28) gave the most number of unfilled grains, and this was followed by Japanese rice (44.54), while Red rice (22.14) produced the least number of unfilled grains and this was followed by the number of unfilled grains of Black rice (30.11).

An interaction effect between the application of bokashi with farm manures and different rice varieties of indigenous rice was also noted; Kaponpon rice applied with any of the bokashi with different farm manures gave a comparable highest number of unfilled grains. While the least number of unfilled grains was obtained from Red rice (21.13) applied with bokashi with sheep manure which also found to be comparable with Black rice

Rice varieties significantly affected the number of unfilled grains. It was sought out that there are more numbers of unfilled grains on the different varieties of rice than those of the bokashi with different farm manures.

Number of Spikelets per panicle

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanes e		
A- Chicken Dung	20.73b	30.40b	56.06a	26.16b	133.35	33.34
B- Sheep Manure	17.80c	25.03bc	62.10a	28.03b	132.96	33.24
C- Hog Manure	23.90b	8.46c	53.13a	28.96b	114.45	28.61
D- Nutriboost (control)	19.03b	25.60b	50.73a	26.76b	122.12	30.53

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 14

TOTAL	81.46	89.49	222.02	109.91		
MEAN	20.37c	22.38bc	55.51a	27.48b		

Analysis revealed that the number of spikelets per panicle was significantly affected by different indigenous rice varieties but not with bokashi with different farm manures. In terms of bokashi with different farm manures, the mean ranged from 28.61 to 33.34 number of spikelets per panicle. On the other hand, the highest number of spikelets per panicle in terms of different indigenous rice varieties was obtained from Kaponpon variety with the mean of 55.51 spikelets per panicle, while the the least spikelets per panicle was obtained from Red rice variety with the mean of 20.37 spikelets.

Moreover, there was a significant difference between different indigenous varieties and bokashi with different farm manures. Using Bokashi with Sheep manure on Kaponpon variety yield the greatest number of spikelets per panicle with the mean of 62.10 spikelets, but analysis suggest that this was comparable to Kaponpon variety using Bokashi with chicken dung, hog manure and nutriboost, with the mean of 56.06, 53.13, and 50.73 number of spikelets.

Number of Spike per spikelets

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanes e		
A- Chicken Dung	10.33bc	9.16c	16.53a	12.76b	48.78	12.10
B- Sheep Manure	9.30c	9.20c	16.83a	13.03b	48.36	12.09
C- Hog Manure	9.60c	3.16d	16.40a	12.50b	41.66	10.42
D- Nutriboost	10.06bc	9.20c	15.10a	12.16b	46.52	11.63
TOTAL	39.29	30.72	64.86	50.45		
MEAN	9.82	7.68	16.22	12.61		

The analysis of variance for the number of spikes per spikelet showed significant results on the effect of the application of bokashi with different farm manures and a highly significant on the effect of different rice varieties. An interaction effect between the factors was also noted.

Kaponpon rice applied with any of the bokashi made from different farm manures had the most spikes per spikelet. It statistically differed from other rice varieties utilized in the study. Black rice applied with bokashi hog manure gave the least number of spikes per spikelets.

1000 seeds weight (grams)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanes e		
A- Chicken Dung	38.35a	32.19b	24.38c	38.74a	133.66	33.42
B- Sheep Manure	38.87a	27.43c	21.93d	35.04b	123.27	30.82
C- Hog Manure	36.22a	34.59a	21.86b	38.11a	130.78	32.7
D- Nutriboost	39.39a	34.81b	21.88c	36.45b	132.53	33.13

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 14

TOTAL	152.83	129.02	90.05	148.34		
MEAN	38.21a	32.25b	22.51c	37.09a		

The analysis for 1000 seeds weight showed that there was a significant effect on the different indigenous rice varieties but not on bokashi with different farm manures. On the bokshi with different farm manures, the mean ranged from 30.82 g to 33.42 g. On the other hand, the heaviest 1000 seeds weight on different indigenous rice varieties was obtained by Red rice variety with the mean of 38.21 g, but was found to be comparable to Japanese variety with the mean weight of 37.09 g. The least weight was obtained from Kaponpon variety.

Moreover, there was a significant difference between indigenous rice varieties and bokashi with different farm manures. Using nutriboost on Red rice variety yielded the heaviest 1000 seeds weight with a mean of 39.39 g, but this was found to be comparable to Red rice using bokashi with sheep manure, chicken dung, and hog manure with a means of 38.87 g, 38.35 g and 36.22 g respectively, Black rice variety using bokashi with hog manure with the mean of 34.59 g and Kaponpon variety using bokashi with chicken dung and hog manure with the mean of 38.74 g and 38.11 g, respectively. The least interaction was obtained from Kaponpon variety using nutriboost.

Grain yield (tons/ha)

BOKASHI SUBSTRATES	RICE VARIETIES				TOTAL	MEAN
	Red Rice	Black Rice	Kaponpon	Japanese		
A- Chicken Dung	0.14	0.12	0.10	0.12	0.48	0.12
B- Sheep Manure	0.13	0.13	0.09	0.12	0.47	0.12
C- Hog Manure	0.14	0.11	0.07	0.11	0.43	0.11
D- Nutriboost	0.14	0.14	0.09	0.11	0.48	0.12
TOTAL	0.55	0.50	0.35	0.46		
MEAN	0.14a	0.13ab	0.09c	0.12b		

The ANOVA for grain yield in tons/ha revealed a highly significant effect of different indigenous rice but a not significant effect in the application of bokashi with different farm manures. Mean ranged from 0.11 to 0.12 tons/ha.

For the effect of different indigenous rice Red rice produced the highest grain yield with 0.14 tons/ha, followed by black rice (0.13 tons/ha) and Japanese rice (0.12 tins/ha). Lowest grain yield was obtained by Kaponpon rice (0.09 tons/ha).

The ANOVA further revealed that there was no significant interaction effect between the bokashi with different farm manures and different indigenous rice. This implies that the two factor did not interacts in influencing the grain yield of different indigenous rice varieties.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 14

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.

CONCLUSIONS

Based on the results of the study, the following conclusions were drawn:

1. Growth and yield of the different indigenous rice, when applied with bokashi with different farm manures, varies significantly.
2. Indigenous rice varieties had a significant effect on the growth and yield of all the parameters gathered except for panicle length.
3. Bokashi with different farm manures was only found to be significantly effective on plant height, biomass, number of productive tillers, number of filled grains, number of unfilled grains, number of spikelets per panicle.
4. Amongst all the bokashi substrates, it was found that Chicken Dung has the highest mean on six parameters, followed by Hog manure and then Sheep Manure.
5. There was an interaction effect between the varieties of indigenous rice and bokashi with different manures except for number of leaves, stem girth, number of tillers, number of non-productive tillers, panicle length, weight of panicle and grain yield in tons per hectare.

RECOMMENDATIONS

The following recommendations are forwarded based on the results of the study:

1. Grow either red rice, black rice, and Japanese rice for higher grain yield.
2. Use any animal manure such as sheep, hog, and chicken manure in making bokashi.
3. Conduct a similar study for further evaluation.
4. Analyze the physical, chemical, and microbial properties of the soil before and after every cropping.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 14

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	Document Type: DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	
		Page	Page 3 of 14